CS-GY 6643 Computer Vision – Project 3, Kaggle Competition Report

Penalty kick direction from pre-contact video

Final submission: `submission_N6.csv` (public 0.83361)

1 Introduction and background

Task. Each input is a broadcast clip of a football penalty that ends about 0.02 s before the kicking foot reaches the ball, plus a metadata row (preferred foot, gender, resolution, frame rate). Two labels are predicted per clip, both from the kicker's point of view: `kick_direction` (left, middle, right) and `goalkeeper_direction` (left, stands in the middle, right). A third label, `goal_outcome`, was removed from the score on 27 October; the file predicts `goal` everywhere, since no model beat that constant. Submissions are scored by $0.7 \cdot$ kick accuracy $+ 0.3 \cdot$ keeper accuracy on 900 test clips: 360 on the public board, the other 540 on the private board.

Why it is hard.

- **The kick is never shown.** The direction must be read from the run-up, the body just before contact and the keeper's first movement. Middle kicks (14%) and standing keepers (5%) are mostly decided at or after contact.
- **Small, blurred players in many views.** Kickers are often about 100 pixels tall, so pose models swap left and right joints. Behind-the-kicker, side-on, high and low cameras change what a left kick looks like.
- **Little labelled data, unseen sources.** There are 1,300 labelled clips from about 136 source videos, and almost every test source is new. Frame rates range from 15 to 60 fps.
- **A noisy public board.** With 360 clips, the public score has a sampling SD of about 0.02 for a fixed model, which is larger than most of the improvements made.

Prior work. Published video models for penalty direction report modest honest results. Freire-Obregón et al. (2025) compare 22 action-recognition backbones on about 640 web clips and reach about 52% three-class and up to 63.9% left/right accuracy. MambaKick (Velesaca et al. 2026) reaches 53.1% three-class with an input that includes frames after the kick. Ranasinghe and Ranasinghe (2025) report 89.4% with pose-guided attention, but on a random split of 755 events with the clip cut closer to contact, which risks leakage between clips of one shoot-out. On the keeper side, Bar-Eli et al. (2007) find that elite keepers stay in the centre on only 6.3% of 286 penalties, close to the 4.8% "stands" rate here. Within the course, the closest material is detection, tracking and optical flow (Questions 2 and 3) and CNN feature representations (Question 1).

Approach. Detect and track the kicker and keeper, measure their bodies with several pose models, fit a camera to the goal, and turn the pose into *label-free* kinematics on the ground plane. Three readers predict the kick: tree ensembles, temporal pose networks, and a pixel model on frozen ViTPose+ tokens of the last frame. A keeper ensemble reads the keeper's motion in the goal plane. This was expected to work because the body just before contact, in particular the plant foot, the hip and shoulder axes and the swing of the kicking leg, carries the side of the kick, and because frozen pretrained features generalise better than models trained from scratch on 1,300 clips.

2 Datasets

2.1 Preprocessing

The data. The 1,300 labelled public clips come from about 136 source videos, and the 900 test clips from 82; only three sources appear in both sets. Clips last 0.3–9.8 s. Most are 1280 × 720 (1,200 of 1,300), with a few at 854 × 480, 640 × 360 and letter-boxed sizes; frame rates are mostly 25, 30, 50 or 60 fps. One test source (31 clips at 15 fps and 640 × 360) has no labelled counterpart. The classes are unbalanced (Figure 1): 607 left, 181 middle, 512 right kicks; 669 left, 63 standing, 568 right keepers. `public_position_clicks.csv` gives hand clicks on the kicker and keeper for 879 public clips; it is used only to train and check the person picker, never as a model input.

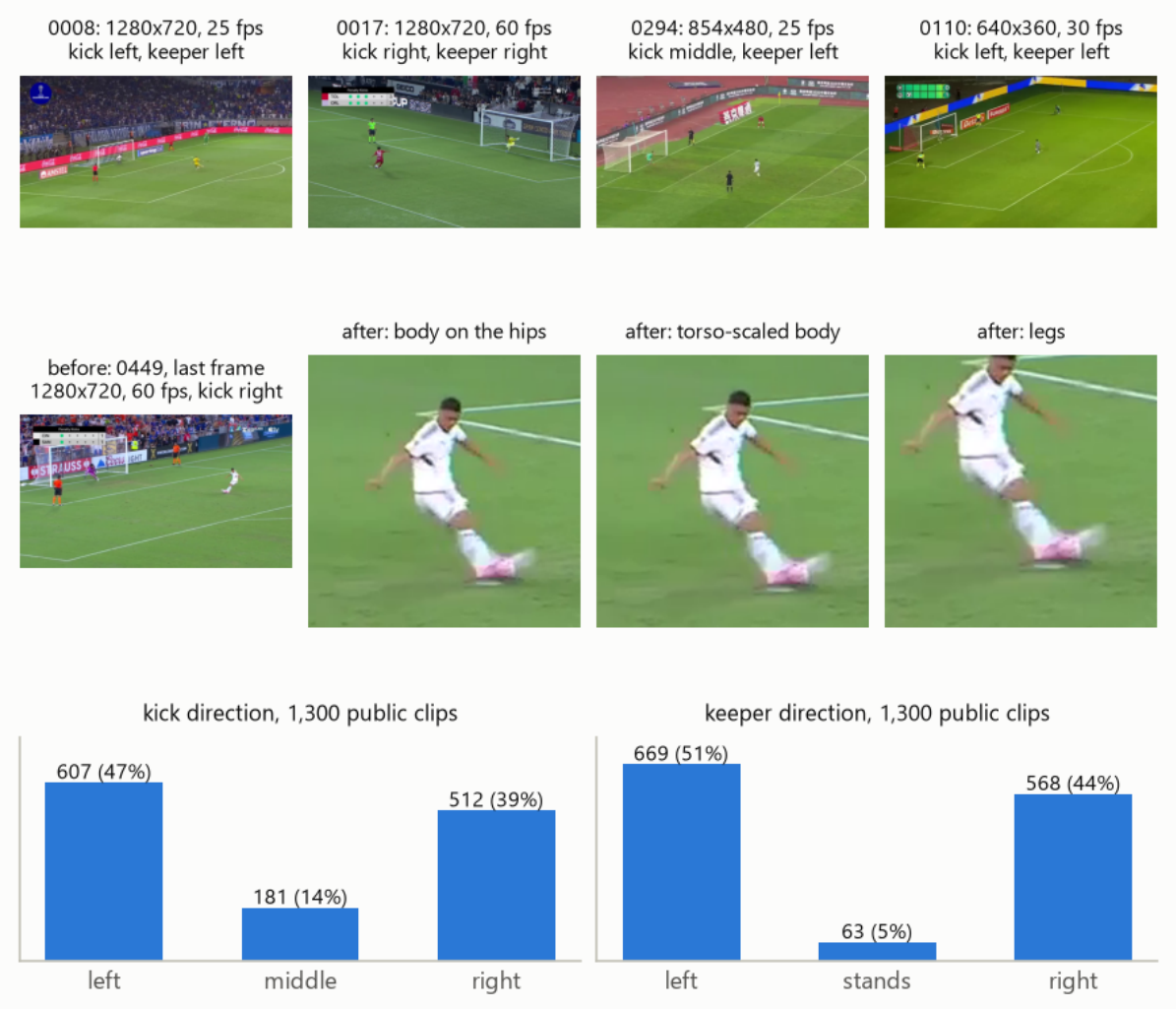


Figure 1: The data and the effect of preprocessing. Top: last frames of four public clips across the resolutions and frame rates in the data; the kicker is often about 100 pixels tall and blurred. Middle: one clip before preprocessing (its full last frame) and after (the three crops the pixel model reads: stabilised to the last frame, centred on the tracked kicker and mirrored so that the goal is on the right). Bottom: the public label distribution; middle kicks and standing keepers are rare.

Steps, in the order applied.

1. **Frame sampling by timestamp.** 13 frames at fixed times from 1.0 s before the end to the last

frame, and every native frame of the last second for dense pose. Time is always handled by timestamp, never by frame index, because the frame rate varies.

2. **Detection.** YOLO11x (Jocher and Qiu 2024) finds persons and the ball on the 13 frames, with a torso-colour descriptor per box. GroundingDINO-tiny (Liu et al. 2024) finds the goal box. Small keepers are missed at full frame, so YOLO11x also runs on an upscaled crop of the goal mouth, with test-time augmentation at confidence 0.05.
3. **Who is who** (Figure 2, top). A learned ranker picks the keeper first, then the kicker given the keeper (features include ball-to-feet distance and goal-box geometry). Trained on the clicks, out of fold by source, it agrees with them on 99.8% (keeper) and 99.5% (kicker) of held-out clips. Each pick is followed over the 13 frames.
4. **Kicker check and re-pick.** 0.02 s before contact the kicker must stand at the ball, on the penalty spot. The *No Bells, Just Whistles* (NBJW) field calibration (Gutiérrez-Pérez and Agudo 2024) projects the spot into the last frame. This label-free check flagged 32 clips (23 public, 9 test) whose track had ended on the referee or another player more than 6 m from the spot; the person within 2 m of the spot is taken instead (Figure 3). A motion-gated track later fixed the kicker identity in 82 clips (53 public, 29 test), and the 46 whose last-frame box moved were re-cropped.
5. **Camera stabilisation.** Frame-to-frame homographies on the background, with the kicker masked, map every frame to the last one.
6. **Pose** (Figure 2, middle and bottom). Five models measure the kicker: RTMW-x whole-body (133 keypoints, with toes and heels), the 2D output of RTMW3D-x (Jiang et al. 2024), and ViTPose++ base, large and huge (Xu et al. 2022, 2024). They are kept as independent *streams*. MotionBERT-Lite (Zhu et al. 2023) lifts the dense ViTPose++ large pose to 3D.
7. **Ball.** Because the clip ends just before contact, the kicking foot must be at the ball. The first pipeline's ball was more than half a kicker height from every foot in 225 clips. Ball candidates from foot-centred crops are scored by support $\times$ confidence $\times \exp(-(d_{\text{foot}}/0.3)^2)$, which moved 244 balls and cut the far-from-feet clips to 24. Where the ball is still weak and more than 1 m from the projected spot, the spot is used (79 clips).
8. **Camera calibration.** Line segments inside the goal box are merged into posts. An orthographic camera with zero roll, elevation θ, azimuth ϕ and scale s is fitted to two known distances: 11 m from the ball to the goal-line centre and 7.32 m between the posts. Image points then map to the ground in metres. Clips without visible posts use the public median elevation; clips whose fit fell below the horizon (12 public, 6 test) get a corrected azimuth.
9. **Label-free features and the mirror.** No feature depends on a pose model's left/right labels: the foot that travels farther is the kicking foot, and body lines are axes taken modulo 180°. Every feature is expressed in a frame where a left/right mirror is exact, so mirrored rows double the training data and serve as test-time augmentation.
10. **Crops for the pixel model.** Three crops of the last frame (body on the hips, torso-scaled body, legs), stabilised, centred on the kicker and mirrored so that the goal is on the right (Figure 1). No resizing beyond each model's input size, colour conversion beyond BGR to RGB, or denoising was applied; contrast normalisation and training on degraded crops were tested and rejected (Section 3.1).

Problem samples and how they are handled. Kickers tracked onto the wrong person (step 4); balls far from every foot (step 7); clips without visible goal posts or with impossible camera fits (step 8); pose left/right swaps (step 9). Nine test clips are frame-identical to nine labelled clips of another source;

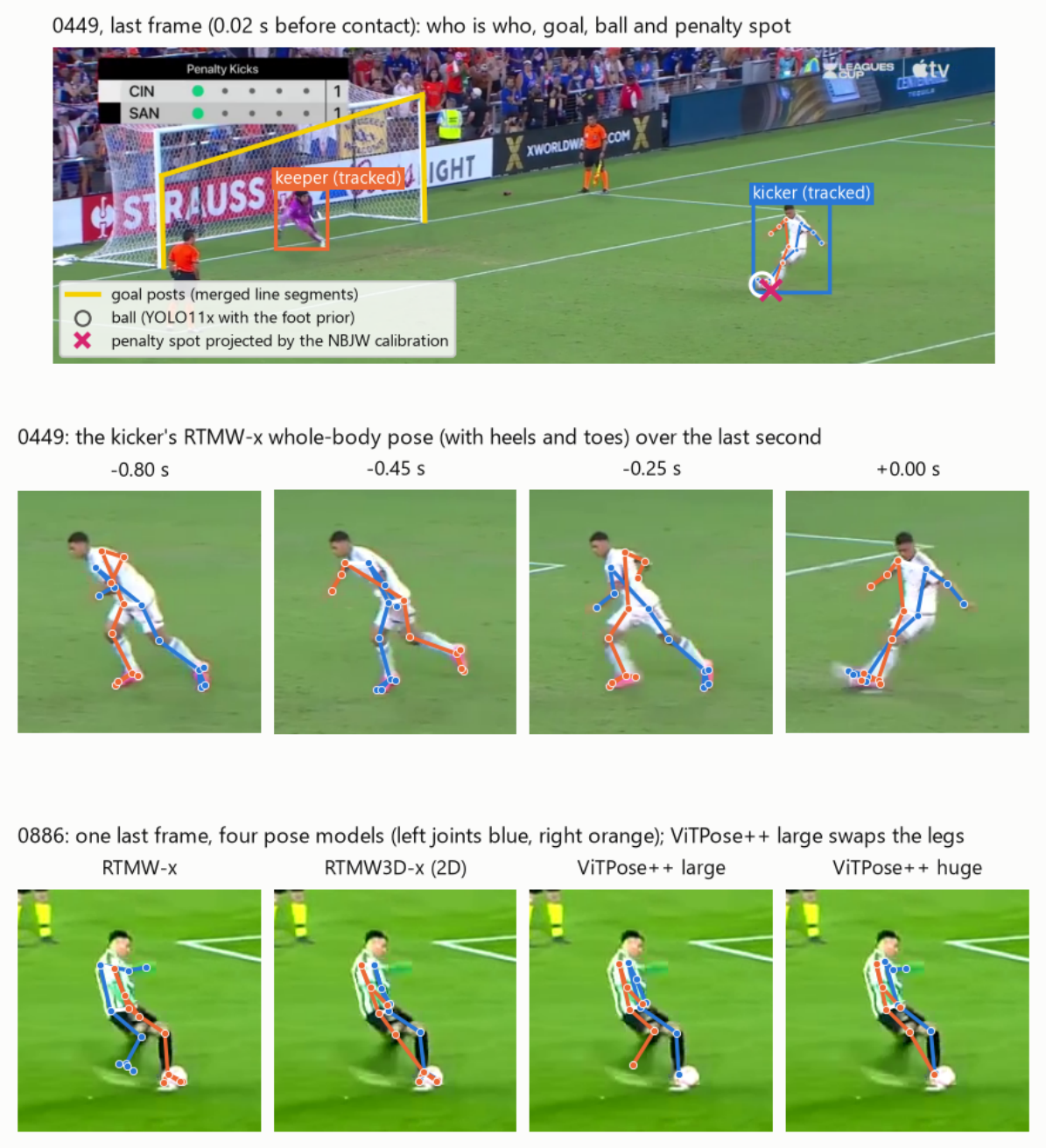


Figure 2: Perception on two labelled public clips. Top: the last frame of clip 0449, 0.02 s before contact, with the tracked kicker and keeper (YOLO11x boxes chosen by the ranker), the goal posts merged from line segments, the foot-prior ball, and the penalty spot projected by the NBJW calibration, which lies under the ball. Middle: the kicker's RTMW-x whole-body pose, including heels and toes, at four times in the last second. Bottom: one last frame read by four pose models, left joints in blue and right joints in orange. ViTPose++ large puts the legs in the same places as the other models but with left and right swapped, which is why every feature is label-free and the pose models are kept as separate streams.

copying their labels would be test-label use, so it was not done. Every test clip is processed on its own: no pooling, normalisation or pseudo-labelling across test clips. Statistics that need many clips, such as the camera prior, come from public clips only.

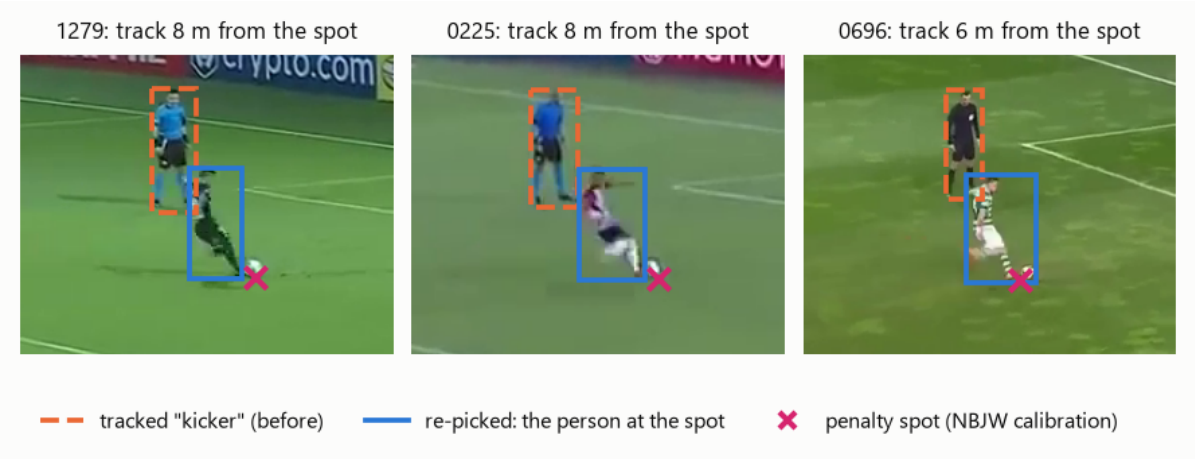


Figure 3: The label-free kicker check on three labelled public clips. 0.02 s before contact the kicker must stand at the ball on the penalty spot. Here the tracked "kicker" (dashed) was the referee, 6–8 m from the spot projected by the field calibration (×). The person at the spot (solid) is taken instead and followed back over the 13 frames. 23 public and 9 test clips were corrected this way; out of fold, these clips had been read at 35% kick accuracy.

Splits. Every number in this report comes from **source-grouped, stratified 5-fold cross-validation** on the 1,300 labelled clips: a model never sees any clip from the source video of the clip it predicts, because clips of one shoot-out share camera, lighting and players and a random split would overstate accuracy. Each evaluation is repeated over three partition seeds (one "set"), and the pooled out-of-fold predictions are scored.

2.2 External and Auxiliary Datasets

No external dataset was used. Two published penalty datasets were vetted: neither had a public release, both had a high risk of overlapping with the competition clips, and one labels directions from the keeper's side. A third holds only collegiate keypoints. The pretrained weights used, all frozen unless noted:

- Ultralytics YOLO11x, COCO (Jocher and Qiu 2024): persons and ball.
- GroundingDINO-tiny (Liu et al. 2024): goal box.
- RTMW-x and RTMW3D-x (Jiang et al. 2024): whole-body pose.
- ViTPose++ base, large and huge (Xu et al. 2022, 2024): pose streams, dense pose, and the pixel model's token maps (base).
- MotionBERT-Lite (Zhu et al. 2023): 3D lifting.
- No Bells, Just Whistles (Gutiérrez-Pérez and Agudo 2024): field calibration and the projected penalty spot.
- WASB (Tarashima et al. 2023): only a foot-reference helper from its code.
- DINOv3 ViT-B/16 (Siméoni et al. 2025): loaded by one script whose metadata file (clip order, flip flags) the final recipe uses.

- ResNet34, ImageNet (He et al. 2016): *fine-tuned* as the keeper CNN.

The model identifiers and checkpoints are listed in the first cell of the submitted notebook.

3 Methods

Formulation. For a clip V (frames with timestamps) and its metadata row m, predict the kick class $\hat{k}$ (left, middle or right) and the keeper class $\hat{g}$ (left, stands or right). Every reader outputs class probabilities $p(k \mid V, m)$ or $p(g \mid V, m)$ and is fitted by minimising a multinomial log loss: the log loss for gradient boosting, the networks and the pixel head (with L2 regularisation), and Gini impurity for ExtraTrees. The prediction is the argmax of the blended probabilities, which maximises the expected accuracy of each label and therefore the metric. Figure 4 shows the pipeline end to end.

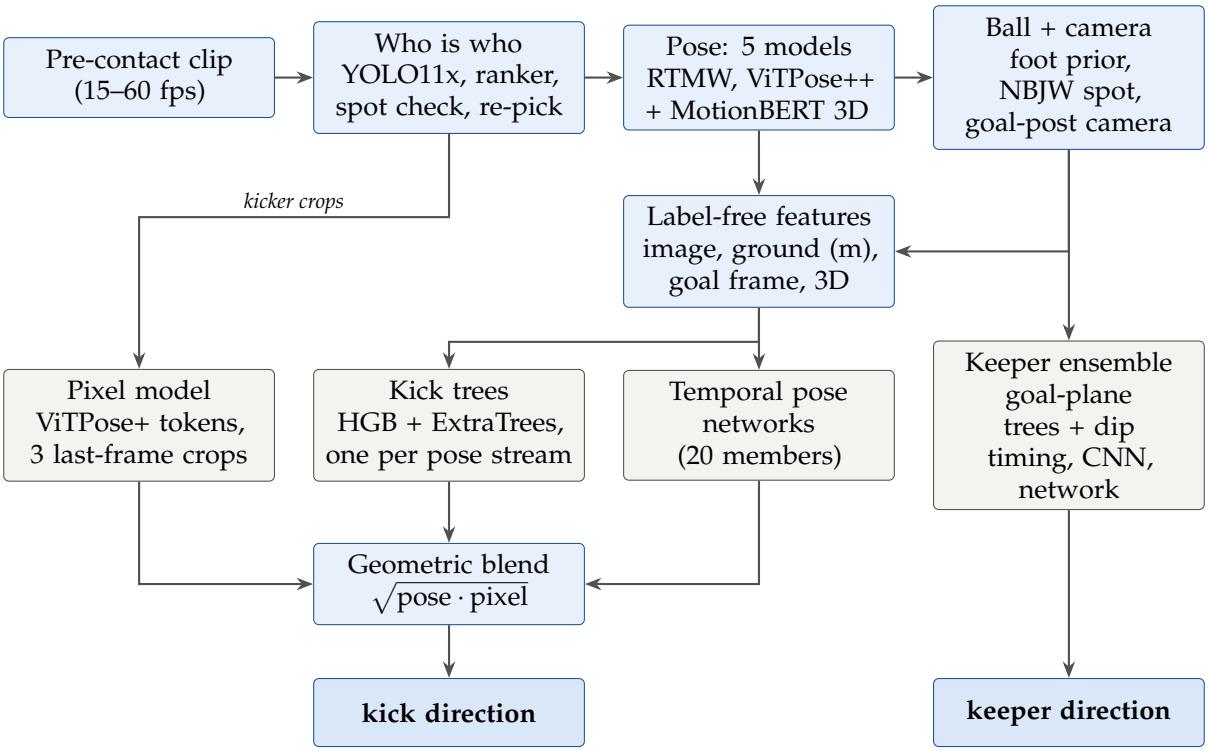


Figure 4: The final pipeline. Blue: shared preprocessing (Section 2.1), the blend and the outputs. Grey: the three readers of the kicker and the keeper ensemble.

3.1 Proposed Method

Features.

- **kin2** (259 features): kicker kinematics in the stabilised, ball-anchored goal frame: run-up angle and speed, the plant and kicking foot relative to the ball, hip and shoulder axes, lean.
- **kinR** (141 features, 36 per-frame channels): the same quantities in metres on the ground. Only what one elevated view can measure is rectified: ground-contact points (feet), same-height differences (hip, shoulder and ear axes) and displacements of roughly constant-height points.
- **kinMB** (29 per-frame channels): the MotionBERT 3D pose in the goal frame, for the networks.
- **Keeper features** (162, plus goal-plane features): the keeper's image motion decomposed on the camera's along-the-goal-line and vertical directions, which is well conditioned in every view. The

final model adds *dip timing* (the time since the keeper's last up/down turning point and the depth of the last dip) and "committed yet?" flags (a lateral move of more than 0.10 m in the last 0.12 s, or feet raised more than 0.10 m).

Kick trees. One model per pose stream (RTMW-x, ViTPose++ large, ViTPose++ huge) on [kin2, kinR]. Each averages a bag of histogram gradient boosting with ExtraTrees (Geurts et al. 2006), both from scikit-learn (Pedregosa et al. 2011), fitted on mirror rows and read with mirror test-time augmentation.

Temporal pose networks. An input projection, three residual dilated 1D convolutions, in the spirit of temporal-convolution pose models (Pavllo et al. 2019), and a head on [last frame, mean of the last six, maximum] plus static features. A clip's pose streams are separate training samples and are averaged at test time, a form of measurement-noise augmentation. The kick network is the mean of 20 members.

Pixel model. This was the largest single gain of the project. A frozen ViTPose+ base backbone turns each of the three last-frame crops into an $8 \times 6 \times 768$ token map. The maps are joined, standardised and reduced by PCA to 512 dimensions. A *camera-aware* logistic head reads them: the features are the tokens times $(1, \cos\phi, \sin\phi)$ times $(1, \text{foot sign})$. Because the camera azimuth ϕ is bimodal, this is in effect a separate linear model for each camera side and kicking foot. Most of the signal is the *motion blur of the swinging leg* in the last frame: in an early test, the left/right AUC of a body crop fell from 0.83 at the last frame to 0.74 at -0.08 s and 0.60 at -0.2 s. Figure 5 shows what it reads.

Blend and keeper. The pose base is $0.5 \cdot$mean(three tree streams) $+ 0.5 \cdot$mean(20 networks). The kick is normalise $\left(\sqrt{\text{base} \cdot \text{pixel}}\right)$. The keeper is $K =$ the mean of seven members ($2\times$ RTMW trees, $2\times$ ViTPose-L trees, RTMW and ViTPose-L goal-plane trees, and a ResNet34 fine-tuned on goal-aligned keeper crops), combined with the temporal keeper network (ten runs) as $(7K + \text{NN})/8$.

Design reasoning and alternatives. Each choice below was decided by source-grouped CV. From 4 November, every candidate change had a rule file written *before* its first result, which fixed the variants, the partition sets, a fresh, never-used decisive seed set, and the thresholds (for example, at least $+2$ clips on the decisive set, $+5$ summed over four sets, and no set below -1). The public board was used only to confirm a CV-chosen file.

Label-free features and separate pose streams.
Joint swaps are common at 100 pixels. Making the five streams agree through a per-frame consensus cost 6–8 clips: the diversity of the members is worth more than cleaner joints.

Ball with a foot prior (25 Oct).
225 clips had a ball far from every foot, which is physically impossible 0.02 s before contact. Re-detecting the ball near the feet, with the spot as a fallback, was the first public gain beyond the first pipeline.

Kicker re-pick (28 Oct).
The 32 clips with the wrong person were read at 35% kick accuracy. Rebuilt trees gained $+7$ net on two seed sets; the networks did not, so only the trees were rebuilt. This choice was made after seeing the CV and is reported as such.

Frozen ViTPose+ tokens (31 Oct).
A frozen DINOv3 crop model jumped once its head knew the camera side and the foot. A sweep then found that ViTPose+ base beats DINOv3, DINOv2 and Sapiens tokens; early fusion of the three crops beats late fusion; and fine-tuning two blocks lost to frozen features (0.818 against 0.887 after 8 epochs) with only 1,300 clips. Shuffled-label controls gave AUC 0.49–0.52 (chance).



Figure 5: What the pixel model reads: the three last-frame crops for four labelled public clips. Rows show its most confident correct left, middle and right calls (out of fold) and its most confident miss of a middle kick (bottom), which looks like a side kick in the last frame.

Crops on the right kicker (1 Nov).
An input audit found that the crops of re-picked clips were still cut on the original track, that is, on the referee. Recutting them gave $+2/+3$ clips.

Geometric blend (4 Nov).
It beat the linear blend by $+2/+3/+3$ clips with lower log loss on every set. A blend gain was always split into true left/right corrections and dropped low-confidence middle calls, because blending plain class frequencies "gains" as much as many pixel blends.

Keeper dip timing (6 Nov).
$+3/+2/+2/+2$ keeper clips on four sets, mostly through more correct "stands" calls.

Tried and dropped: follow-through and swing-direction cues (AUC ≤ 0.55); ball direction read from the last frames; foot pointing; run-up rhythm ($+4$, below the bar); head tilt and foot shape (-5); middle-kick stackers (-3 to -7); Sapiens pose; DINOv2, DINOv3 and MViT pixel features; higher-resolution crops (-3, -39); other crop alignments (-11); contrast normalisation (-6 to -8); training on degraded crops; a 15 fps time grid; scaling P(middle) by 0.8 (-5); a sharpened pixel model (-7); LightGBM; stackers; a separate middle model; a keeper pixel model; keeper "stands" reweighting; and an outcome model.

4 Results

4.1 Training Details

- **Validation.** Source-grouped, stratified 5-fold CV (Section 2.1), repeated over partition seeds. Partition seeds 0–26 were used up by the pre-written rules, each new rule taking the next unused ones; the final model is reported on four sets: seeds 0–2, 3–5, 18–20 and 21–23.
- **Final values.**
  - Kick trees: histogram gradient boosting, 4 configurations $\times$ 3 seeds, averaged with ExtraTrees (500 trees, minimum leaf 3, $\times$ 3 seeds).
  - Networks: AdamW (Loshchilov and Hutter 2019) with a one-cycle schedule, label smoothing 0.1 and mirror augmentation; two channel sets (with and without the ground-plane and 3D channels) $\times$ two stream groups $\times$ five configurations, each over three seeds.
  - Pixel head: PCA to 512 dimensions and logistic regression with $C = 0.001$.
  - Blend weights: 0.5/0.5 for the pose base, then the geometric mean; the keeper uses $(7K + \mathrm{NN})/8$.
- **What was searched.** Pose-model sets and stream fusion; ball, camera and crop variants; pixel backbones (ViTPose+ base, DINOv3, DINOv2, Sapiens, MViT); crop sets and resolutions; early versus late fusion; frozen versus fine-tuned backbones; linear, geometric and stacked blends; class-probability scaling; and the alternatives listed in Section 3.1. The exact grids are in the submitted code.
- **Hardware and runtime.** One laptop: NVIDIA RTX 5070 Ti Laptop GPU (12 GB), 24 CPU threads, 34 GB RAM, Windows 11, Python 3.14, PyTorch 2.14 with CUDA 13.0. The heavy perception stages (detection, five pose models, dense pose, 3D lifting, calibration, crops, token maps) took many GPU-hours over the project and are cached; they were not re-timed end to end. The final run of the notebook, which recomputes every CPU stage (features, all tree and keeper fits, the pixel head, every CSV step) from the cached perception, took 79 minutes on 6 November.

- **Reproducibility.** The notebook rebuilds `submission_N6.csv` byte for byte. Each script is written by a `%%writefile` cell into a clean-room folder and run there, and all seven CSVs of the chain (P4 to N6) came out identical to the submitted ones. The CPU fits are seeded and deterministic; GPU network training is not bit-for-bit deterministic, so retraining the networks from scratch can change a few test predictions.

4.2 Results

Partition set (seeds)	0–2	3–5	18–20	21–23	mean $\pm$ SD
Kick right (of 1,300)	996	1006	994	995	997.8 ± 5.6 (76.8%)
Keeper right (of 1,300)	1203	1199	1198	1200	1200.0 ± 2.2 (92.3%)
Metric	0.8139	0.8184	0.8117	0.8127	0.8142 ± 0.0030

Table 1: The final model (N6), source-grouped 5-fold CV on four partition sets. The SD is across partition sets, that is, across fold assignments of the same 1,300 clips.

Validation and the public board. Table 2 lists each main version with its CV and public scores; it doubles as the ablation, since each row adds one component to the row above. The public board scores N6 at 0.83361, against 0.52277 for the "educated guess" baseline (kick and keeper to the kicker's natural side). The public score sits 0.02 above the CV level, which is 1.7 SD of the 360-clip sampling noise. No overfitting is needed to explain it: the random 360/540 split alone gives SD(public $-$ private) $= 0.022$ for any model. Since $900 \times$ true level $= 360 \times$ public $+ 540 \times$ private, a lucky public board leaves a lower expected private score, forecast at 0.820 ± 0.017 (80% range 0.798–0.842); this forecast method held its stated 80% and 95% ranges 80.3% and 95.4% of the time on 20,000 pseudo-boards. N6 and N4 make identical kick predictions and differ on 7 of 900 test keepers: out of fold N6's keeper won the disputed calls 13 to 4, while on public it lost 3 clips (0.83361 against 0.83611). The expected private difference is about -0.0002, a tie, and N6 was kept by the rule written before any test feedback.

File	Component added (CV on seeds 0–2)	Kick right	Keeper right	CV metric	Public
v10	First pipeline: YOLO11x, RTMW / ViTPose++ pose trees, temporal networks, keeper CNN	927 (71.3%)	1187	0.773	0.77708*
P4	Foot-prior ball, ground-plane camera, goal-plane keeper, MotionBERT channels	–	–	–	0.81472
P4rkT	+ spot ball, keeper network, kicker re-pick: *the pose readers*	953 (73.3%)	1200	0.790	0.81861
–	*The pixel model alone* (kick only)	74%	–	–	–
P4rkD	+ pixel model, linear blend, crops on the right kicker	994 (76.5%)	1200	0.8122	0.83611
N4	+ geometric blend, corrected camera for 6 test clips	996 (76.6%)	1200	0.8132	0.83611
N6	+ keeper dip timing and "committed yet?" flags	**996**	**1203**	**0.8139**	0.83361

Table 2: Main versions and ablation. Each row adds one component. The pose readers alone (73.3%) and the pixel model alone (74%) are each well below their blend (76.6%). *Old three-label formula; P4 was not cross-validated as a whole.

Secondary metrics. Out of fold, the pixel model alone reaches a left/right AUC of 0.92, and its learning curve gains about 2.8 accuracy points per doubling of the labelled clips. Kick accuracy rises with confidence, from 0.55 in the lowest fifth to 0.92 in the highest (Figure 7). Figure 6 shows the confusion matrices: left and right kicks are read at 90% and 86% recall, middle kicks at 8%.

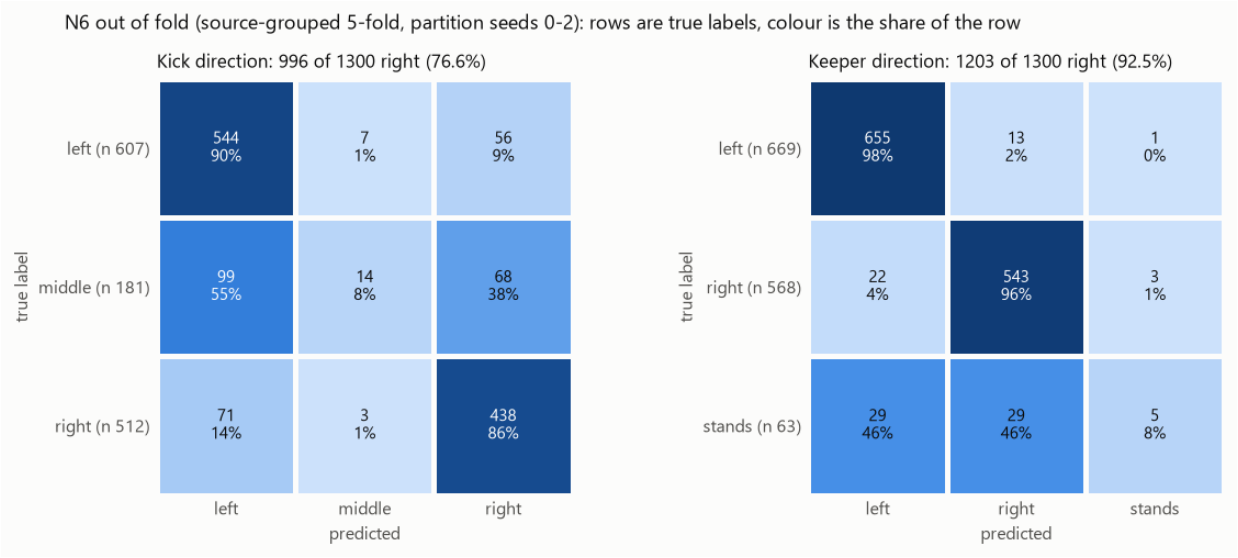


Figure 6: The final model out of fold (seeds 0–2), rows normalised. Left and right kicks are read well; middle kicks are mostly called as sides, and keepers who stand are mostly called as divers.

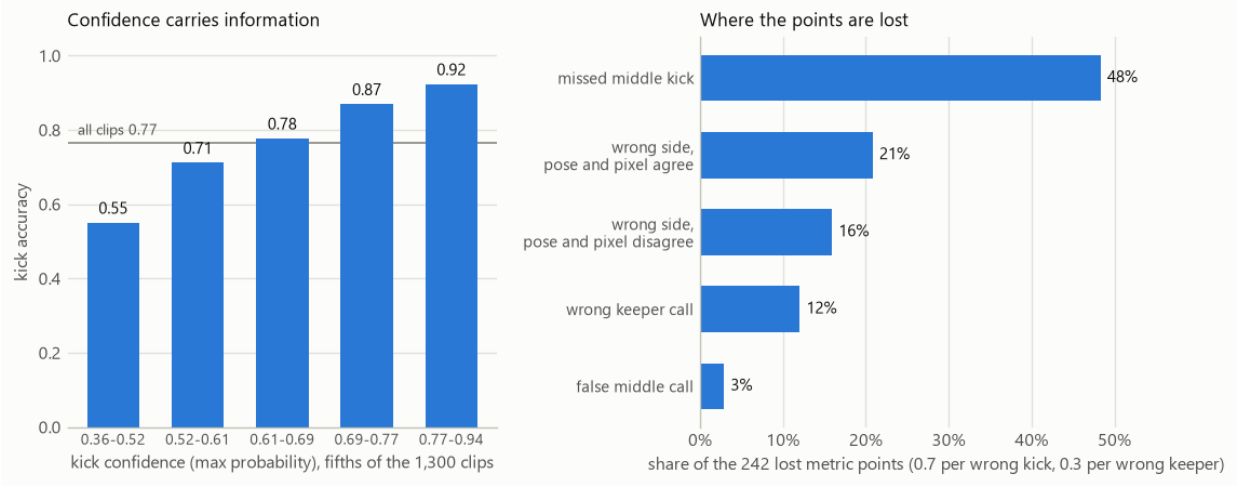


Figure 7: Left: kick accuracy by confidence fifth. Right: where the 242 lost metric points go (0.7 per wrong kick, 0.3 per wrong keeper).

Where the points are lost. 48% are missed middle kicks; 21% are wrong-side kicks where the pose and pixel models agree and are both wrong; 16% are wrong-side kicks where they disagree; 12% are keeper errors, mostly missed "stands"; 3% are false middle calls. Every component ranks middles at an AUC of only 0.61–0.66.

Is the model guessing? No. With the predictions fixed and the labels reshuffled 100,000 times, the metric is 0.443 ± 0.010 against 0.814, 38 SD away (34 SD when reshuffling within each source). Retrained on shuffled labels, the kick trees fall from 0.689 to 0.42–0.45. Against the natural-side baseline, N6 wins 396 kick clips to 52 (McNemar $p \approx 10^{-67}$), is ahead in every subgroup (fps, resolution, foot, gender, camera side, class), and leads on all of 100,000 simulated 540-clip private boards, with a smallest margin of $+0.19$.

5 Use of AI

- **Tool and model.** Claude Code (Anthropic's coding agent) with the model **Claude Opus 5.5**, in one main session from 23 October to 8 November 2026. The first pipeline (v10 in Table 2; Sections 1–5 of the notebook) was built in an earlier Claude Code session (23–25 October) that ran mainly on Claude Sonnet 5.5.
- **Exact prompts.** All of the author's prompts, verbatim and time-stamped, with the assistant's replies and a line per tool call, are in the chat histories submitted with the course hand-in (not published here). Representative prompts from the main session:
  - 23 Oct: "focus only on Q4, fully absorb it, and its dataset [...] ill ask you questions later"
  - 23 Oct: "the goal is to be #1 on the private leaderboard, which are all clips never seen, so the goal is to create a model that generalizes the best"
  - 23 Oct: "dont start building anything until i tell you to, i am just ideating here"
  - 23 Oct: "my first goal is to create more data, i think if i divide the entire screen into axes (X and Y), mirror each pixel, it should create double the data."
  - 6 Nov: "do a final comprehensive assessment that can say for certainty that our model is not guessing, is always better than chance.csv on the leaderboard [...]"
  - 8 Nov: "draft a detailed report overviewing the entire pipeline, its main parts and the decision making process(.tex file)"
- **Rationale.** The pipeline spans 2,200 clips, a dozen pretrained models and hundreds of controlled experiments. The assistant was used to write and run that code, search the literature, and keep the experiments honest (source-grouped CV, rules written before results), within constraints the author set: ideas discussed before anything was built, read-only data, resource limits on the laptop, and no submission without the author's approval.
- **Output and how it was used.** The code in the submitted notebook and scripts was written and run by the assistant at the author's direction. The author set the goals and constraints, proposed directions (for example the mirror augmentation above, which became the exact left/right mirror of the features), reviewed results, made every Kaggle submission and chose the final file. This report was drafted by the assistant from the project's logs and revised at the author's request.

6 Conclusions

What we learned. Validation that matches the test (unseen sources) mattered more than any single model: random folds and the 360-clip public board both mislead. Physical constraints made cheap,

label-free checks: the kicker must stand at the ball on the spot, and the foot must touch the ball. These found errors that no model could. The strongest single cue was not in the pose at all but in the pixels: the motion blur of the swinging leg in the last frame. Frozen pretrained features beat fine-tuning at 1,300 clips.

Limitations. Middle kicks (8% recall) and standing keepers are mostly decided at or after contact, which the clips do not show; they account for more than half of the lost points. The heavy perception stages are cached rather than re-run in the notebook, and GPU training is not bit-for-bit deterministic. The private score is expected near 0.82, below the public 0.834, because of split noise rather than overfitting.

Going forward. The pixel model's learning curve (+2.8 points per doubling) makes more labelled clips the strongest lever, if an external dataset without overlap became available. Other steps would be a keeper model that reads commitment timing at higher temporal resolution, and a pixel model on the last two or three frames to capture the blur direction.

7 Author contributions

The author is the only team member and did all parts of the project: problem framing, direction of the experiments, review of results, Kaggle submissions and the final selection, with the AI assistance described in Section 5.

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